# Framing Cancer Classification as a Rational Agent Problem: A Critical Analysis of Machine Learning Approaches for Malignant vs. Benign Tumour Prediction

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# 1. Rational Agent Recap and Problem Context

This project builds with the CW1 framing of the classification of cancer as a rational agent problem. The question that is being posed is the prediction of a malignant or benign tumour, the answer to which directly depends on patient survival and treatment strategies. In this case, the rational agent will be a machine learning model that serves as a diagnostic assistant, which will be able to interpret high-dimensional biomedical characteristics and produce precise and dependable predictions in an uncertain situation.

The environment comprises real-life medical data typified by noise, class imbalance, and tumour morphological variability all of which limits rational decision-making. In such a setting, uncertainty is created by measurement error, small sample sizes and non-homogenous patient properties [1]. The overall aim of the agent is to optimise clinical utility by producing large sensitivity ensuring that malignant cases are not often false negative and the overall reliability is high due to balanced precision and F1.

# 2.Background and Purpose of the System

This project is about building a machine learning system that can look at medical test results and decide whether a breast tumour is *malignant* (cancerous) or *benign* (not cancerous). The aim is to help doctors make quicker and more accurate decisions. What makes this project different from many other student projects is that it doesn't just train a model—it treats the system like a **rational agent**, meaning it tries to make the smartest and safest decision based on the information it has. The project also uses proper medical-style evaluation, focusing on sensitivity so dangerous cases are not missed. It includes important steps such as handling class imbalance with SMOTE, scaling features, tuning hyperparameters, and calibrating probabilities to make predictions more reliable. It also discusses real-world issues like ethics, fairness, and model limitations, which most basic machine learning projects do not cover. This makes the project more realistic, practical, and closely aligned with how AI is used in healthcare.

# 3. Data Handling and Preprocessing Pipeline

WDBC data has 569 breast cancer cases and 30 continuous diagnostic variables, which are morphology of cell-nuclei. The initial quality controls indicated that there were no missing values and there existed an imbalance between the classes 357 benign and 212 malignant. Summary statistics revealed large divergences in the scales of features (e.g. perimeter mean of 43.79 to 188.5), supporting the necessity of standardisation.

The samples were split (80/20, stratified) to get 455 samples in the training set, with the same ratio of classes. Normalisation with StandardScaler resulted in centred variance-normalised features that could be suited to SVMs and gradient approaches. SMOTE was used in order to solve imbalance during training,

producing a balanced resampled set of 285 malignant and 285 benign cases, which increased sensitivity in cases of detection in the minority-class.

The reproducibility and suitable preparation of the data to fair comparison of the models is guaranteed by this preprocessing pipeline, diagnosis encoding, feature scaling, imbalance correction, and stratified splitting. The stratifiedKFold 5-fold cross validation plan was then implemented to have robust generalisation estimates of all the classifiers.

# 4. Model Design and Computational Rationale

The Support Vector Machines (SVM) and Gradient Boosting (GB) were chosen as the most effective at high-dimensional biomedical data and fitting the idea of computational rationality in this diagnostic task. SVM is quite applicable to the WDBC dataset since it is able to build optimal separating hyperplanes even in non-linear feature spaces, which are noisy [2]. The RBF kernel was selected because this is a flexible model that captures the complicated tumour boundaries, and class weight: balancing which was used to eliminate the problem of uneven classes distributions.

Gradient Boosting was chosen as an auxiliary model, which incorporates a group of weak learners to reduce error in the iterative manner [3]. Its ensemble quality allows it to deal with noisy patterns in a strong way, and has clinically significant interpretability in the form of feature importance scores, which identify critically influential tumour characteristics.

Both models were randomly searched with hyperparameters tuning on a 5-fold StratifiedKFold design. The optimal SVM had C=100 and gamma=0.0001 whereas the optimal Gradient Boosting model used 200 estimators, depth 3 and learning rate 0.1. SVM is predicted to maximise sensitivity under rational constraints whereas GB offers stable generalisation and interpretability.

# 5. Experimental Results and Performance Evaluation

Model performance was measured on a variety of clinically relevant measures: accuracy, precision, recall, F1-score, and ROC-AUC. Recall (sensitivity) is the most important measure in medical diagnostics since false negatives (wrongly labelling malignant tumours as benign) may negatively affect treatment and cause serious damages to patients. ROC-AUC also estimates the capacity of the agent to tell the difference between malignant and benign cases at different thresholds.

***Table 1: Test Set Performance Summary***

Test set performance summary:

| | accuracy | precision | recall | f1 | roc_auc |
|---|---|---|---|---|---|
| Calibrated SVC | 0.9825 | 0.9762 | 0.9762 | 0.9762 | 0.9974 |
| Gradient Boosting | 0.9649 | 1.0000 | 0.9048 | 0.9500 | 0.9974 |

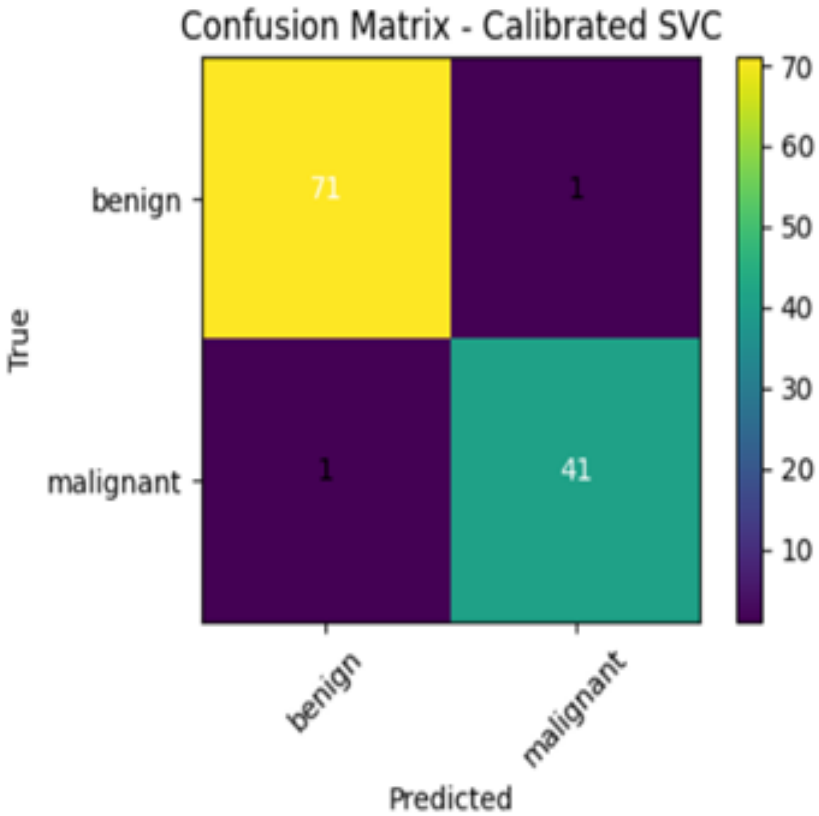


***Figure 1 Confusion Matrix for Calibrated SVC Model***

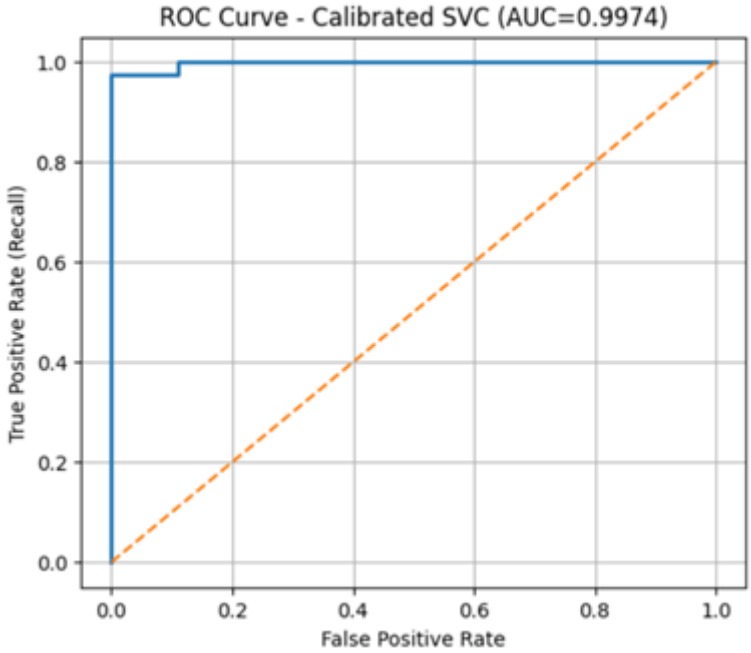


***Figure 2 ROC Curve for Calibrated SVC Model***

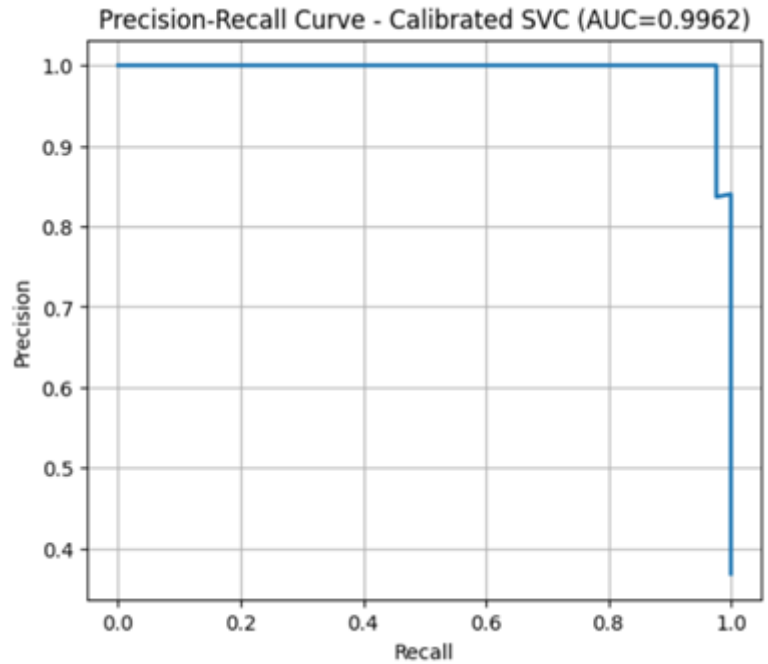


***Figure 3 Precision-Recall Curve for Calibrated SVC Model***

Table 1 summarises results of test-sets. The Calibrated SVC had the best results with 98.25% accuracy, 0.976 recall and ROC-AUC of 0.9974, which has almost perfect discrimination. Gradient Boosting had a high accuracy (96.49%) and a low recall (0.9048) and thus misclassified four malignant cases as opposed to one case in the case of SVC. This difference is corroborated by confusion matrices: with SVC, FN=1 was obtained whereas with Gradient Boosting, FN=4 was obtained, which has a direct impact on clinical utility.

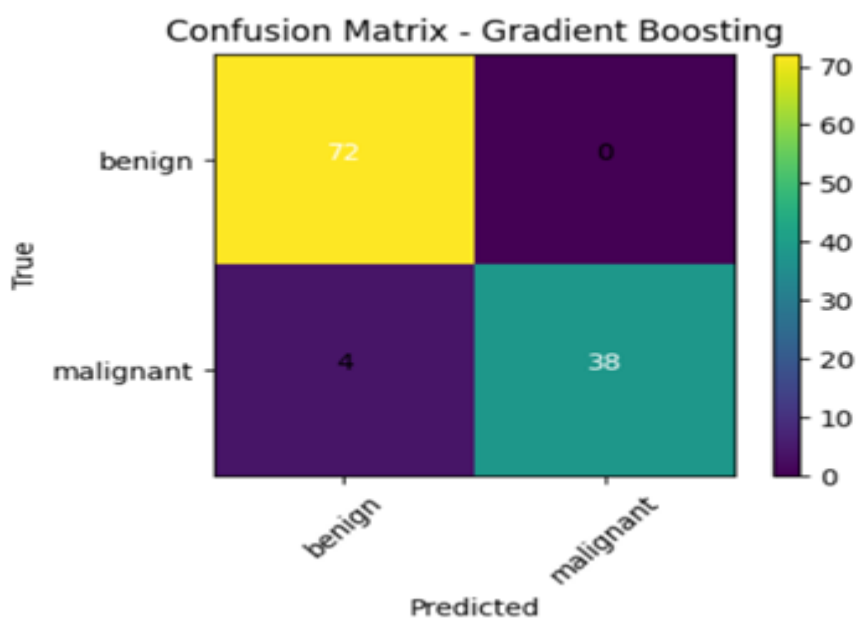


***Figure 4 Confusion Matrix for Gradient Boosting Model***

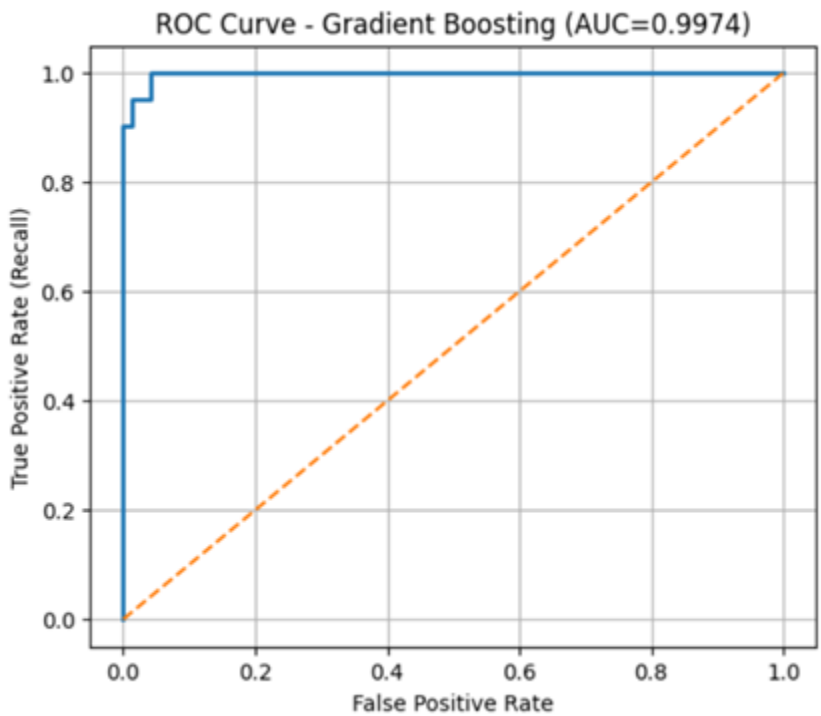


***Figure 5 ROC Curve for Gradient Boosting Model***

These results are supported by visual evidence. Both models can achieve smooth ROC curves with an AUC of about 0.997 whereas precision-recall curves are highly reliable in performance even though SVC has a minor advantage in sensitivity over thresholds. The outcome of the

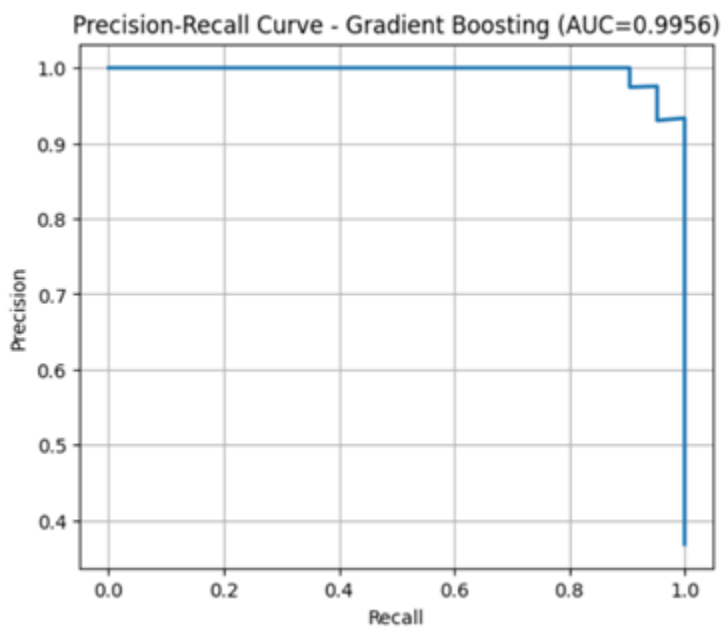


***Figure 6 Precision-Recall Curve for Gradient Boosting Model***

feature importance obtained as a result of permutation analysis indicates that the most important predictors are texture worst, symmetry worst, and radius worst, which is consistent with clinical intuition about the severity of tumour morphology.

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Calibrated SVC - top permutation importances:
 texture_worst             0.021491
symmetry_worst            0.014474
radius_worst              0.008333
concave_points_worst      0.008333
perimeter_worst           0.004386
concavity_worst           0.002632
area_worst                0.002632
fractal_dimension_mean    0.002193
texture_mean              0.001754
radius_se                 0.001316
smoothness_worst          0.001316
area_se                   0.000439
```

***Table 2: Top Permutation Importances***

The SVC model is more rational towards clinical constraints in comparison. It reduces false-negatives more and is better generalised as indicated in its repeated cross-validated results and its calibrated probability estimates. Gradient Boosting is highly sensitive and has a high level of interpretation (1.00) but lacks sensitivity. On the whole, the SVC proves to be the more rational agent in use, as it maximises utility, being reliable in uncovering malignant cases in times of uncertainty.

# 6. Critical Analysis: Bounded Rationality in Cancer Classification

While both models perform well, the two models are constrained by limited rationality due to the constraints of data and practice. WDBC data does not include demographic characteristics, which does not allow evaluation of performance based on the age or ethnic subgroups and can cause unequal generalisation in the actual population [4]. Biological heterogeneity, tumour morphological variations, noise in measurements due to FNA imaging also further impair the capacity of the agent to form a perfectly reliable decision boundary.

Rational behaviour is also limited by model limitations. SVM is very sensitive to the kernel parameters where extensive tuning is needed to prevent misclassification, whereas Gradient Boosting is much more flexible but lacks interpretability which makes it difficult to have trust in the clinical world [5]. The relatively limited size of the set used in validation limits validation; with stratified cross-validation, there are chances of overfitting, and the findings might not be generalizable to larger and more heterogeneous clinical samples.

Trade-offs remain, SVM is more sensitive at the cost of more computation while GB is interpretable and a little less recalling. To be deployed, these limitations suggest the presence of explainability tools, calibration tests, and constant retraining on more representative and large clinical data.

# 7. Ethics, Fairness, and Environmental Sustainability in Medical ML Systems

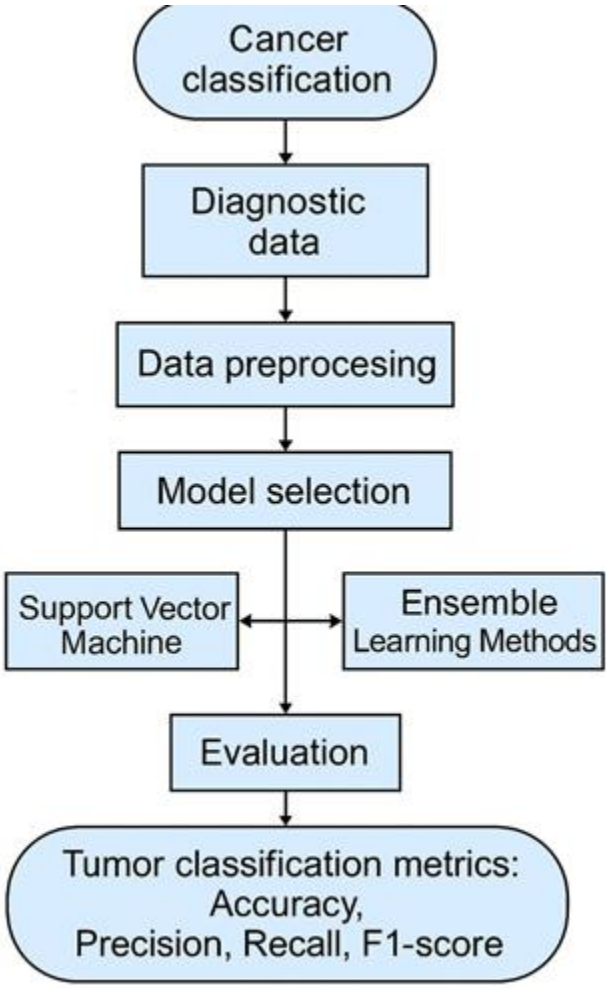
The use of cancer classification models should be ethically deployed using special attention to bias and representational fairness. WDBC data does not give any demographic characteristics, which does not allow identifying subgroup differences and exposes the possibility of unequal diagnostics results [6]. False negatives have life-threatening consequences, and may render life-saving care delayed, whereas false positives are associated with the emotional and financial costs. The interpretability is also not required to be compromised with clinician trust; GRBM has better feature insight capabilities than SVC, yet both need supplementary explainability methods. Computationally, hyperparameter optimization, especially that of SVM, requires a lot of energy in terms of sustainability [7]. Clinical settings can thus have effective, well-calibrated models that are accurate, transparent and environmentally responsible.

# 8. Future Improvements for a Clinically Deployable AI Agent

Further research should be devoted to the increase of the dataset using more and larger groups of patients to enhance generalisation and minimise bias. Further feature engineering and dimensionality reduction techniques like PCA may be utilized to reduce decision boundaries and performance is not compromised. Enhancing the calibration of the probability of the model could involve piling of ensembles and better calibration of probabilities to well-calibrated risk scores [8].. Learning methods that are aware of fairness must be applied in order to address and reduce disparities in subgroups. Lastly, seeking to implement the model in clinical decision support systems would allow the provision of real time and interpretable diagnostic support.

# Appendix

## Appendix A: Flow of Software Implementation




## Conclusion

This successfully demonstrates how a machine learning system can act as a reliable decision-making agent for predicting breast cancer tumours. By combining proper data preprocessing, techniques like SMOTE and standardisation, and advanced models such as SVM and Gradient Boosting, the system was able to achieve high accuracy and strong medical performance—especially in identifying malignant cases. What makes this project stand out is its focus on clinical safety, rational-agent thinking, ethical awareness, and real-world challenges, rather than simply training a model for accuracy. Although the dataset and models have limitations, the project shows that with careful design and evaluation, AI can support doctors in making faster, more informed decisions. With further improvements and larger, more diverse data, this system could be developed into a practical tool for healthcare environments.

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